

HIKNet in PyTorch

Classifying head impacts vs false positives from instrumented mouthguards

Guru R Bharadwaj (PES1UG24CS177)

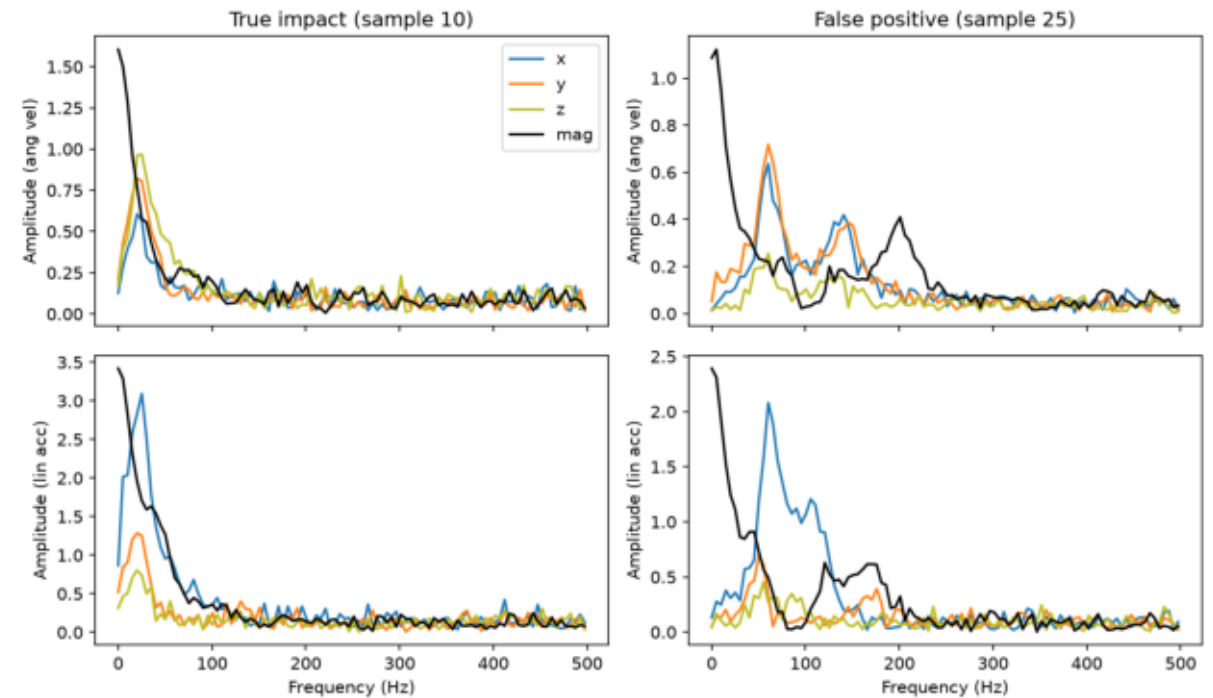
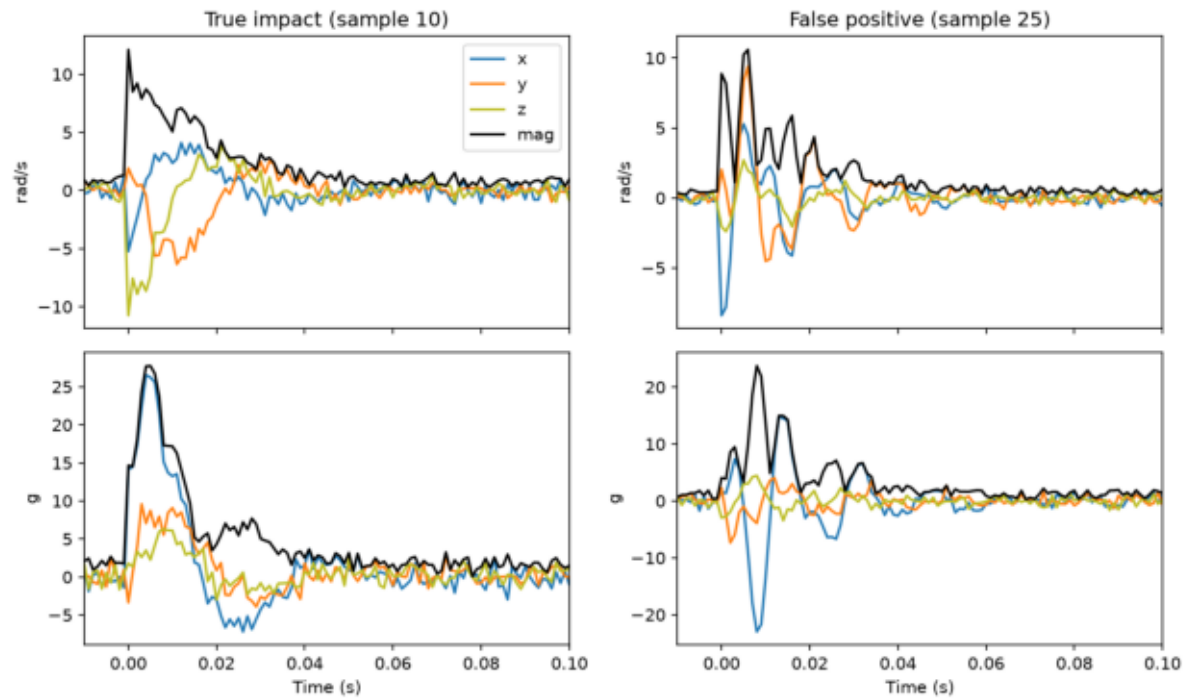
Problem

- Instrumented mouthguards record head kinematics to study head impacts
- Many recorded events are false positives (chewing, talking, dropping the guard)
- Goal: binary classifier - real head impact vs false positive
- Input: 6 channels (linear + angular acceleration x/y/z), 199 samples at 1000 Hz
- Compare a feature-based SVM baseline with HIKNet, a 1D/2D CNN on raw signals

- Synthetic mock dataset: 527 samples (264 impacts / 263 false positives)
- Generated by scripts/make_mock_data.py - the real lab data was unavailable
- Same shape as the original: 6 channels x 199 timesteps @ 1000 Hz
- Signals standardized; stratified 70/30 train/eval split (seed 0)
- Results validate the pipeline only - they are NOT real-world performance

All numbers in this deck come from synthetic data.

Exploratory data analysis



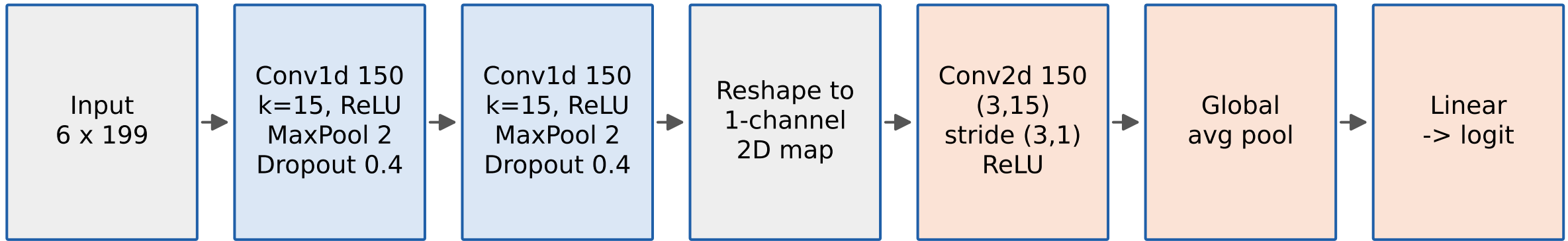
SVM baseline

- Hand-crafted features: PSD bands, wavelet (WT) bands, biomechanical model terms
- Greedy forward selection by CV ROC AUC -> 10 features

#	Selected feature	CV AUC
1	lin acc y PSD 10Hz	0.923
2	ang acc y WT 30Hz	0.960
3	lin acc z PSD 10Hz	0.969
4	lin acc y PSD 130Hz	0.971
5	biomech model 37	0.975
6	biomech model 17	0.975
7	lin acc y WT 120Hz	0.977
8	lin acc y PSD 150Hz	0.978
9	lin acc y PSD 40Hz	0.978
10	lin acc z WT 4	0.978

Metric	CV	Test (32)
Accuracy	0.902	0.969
Precision	0.968	1.000
Specificity	0.983	1.000
Sensitivity	0.782	0.938
ROC AUC	0.978	1.000
PR AUC	0.970	1.000

HIKNet architecture



- Two Conv1d blocks learn temporal filters; Conv2d then scans the (time x filter) map
- GAP head keeps parameters low; sigmoid on the logit gives $P(\text{impact})$

Training

- Optimizer: Adam, learning rate 0.001, batch size 32
- Loss: binary cross-entropy on the logit (BCEWithLogits)
- Early stopping on validation loss, patience 5; best checkpoint restored
- Final run stopped at best epoch 12
- Tuning: each config repeated 10x with different seeds, 30 epochs max

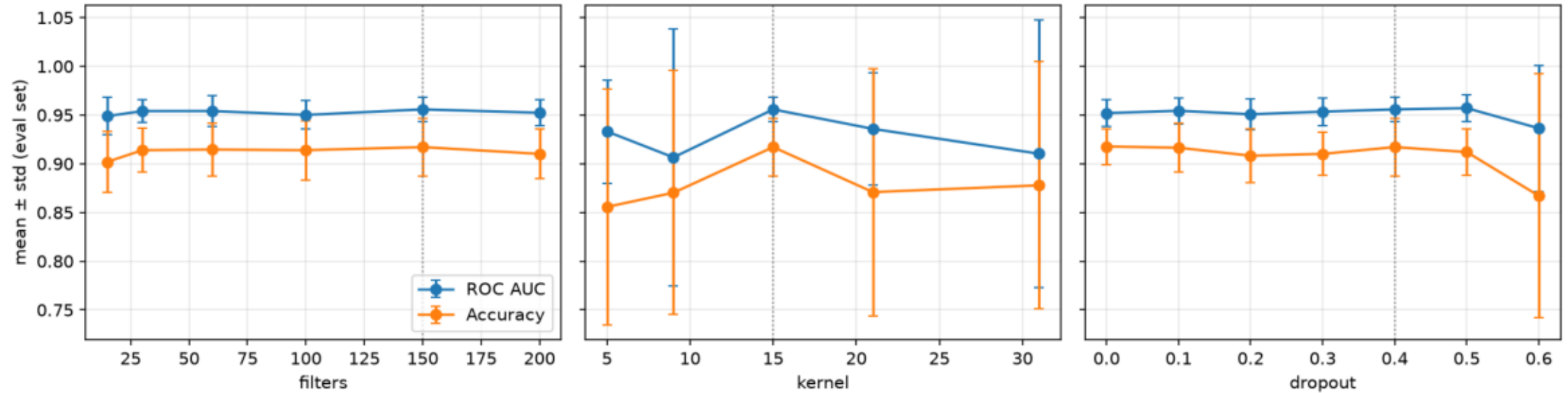
RecursiveNet vs HIKNet: 10-fold CV

Model	Accuracy	Precision	Specificity	Sensitivity	ROC AUC	PR AUC
HIKNet	0.936	0.915	0.909	0.962	0.967	0.961
RecursiveNet	0.820	0.758	0.848	0.793	0.882	0.902

- Same 10 stratified folds for both models; mean over folds
- Higher mean ROC AUC: HIKNet

Tuning: output head and hyperparameters

HIKNet one-at-a-time hyperparameter sweeps (dotted = default)



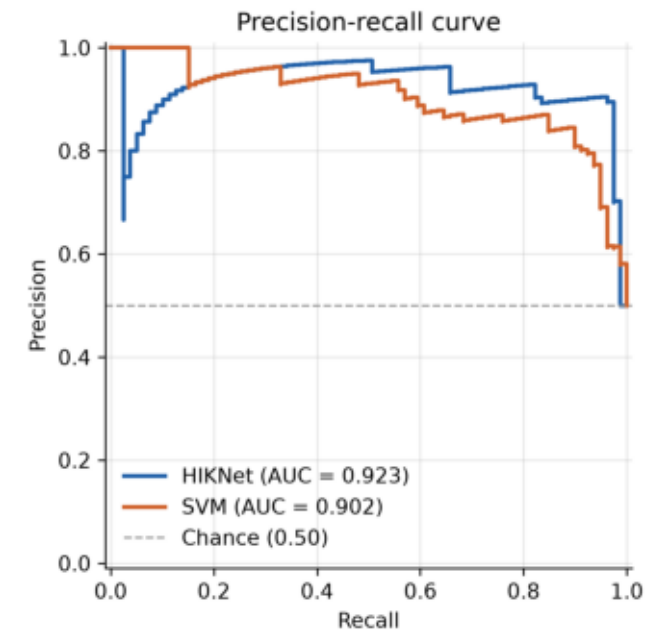
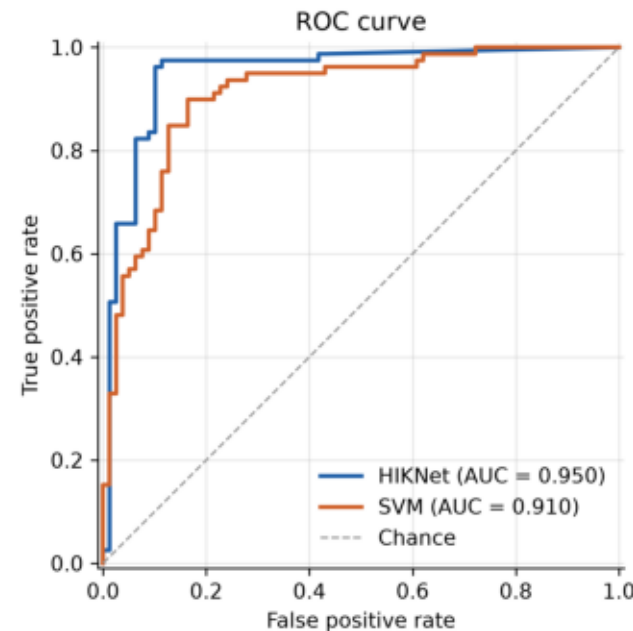
Head	Accuracy	ROC AUC
GAP	0.917 ± 0.030	0.956
GMP	0.894 ± 0.031	0.943
DENSE	0.684 ± 0.186	0.717

- Best head: GAP
- Default: 150 filters, k=15, dropout 0.4
- Best: 150 filters, k=15, dropout 0.5
 - best kernel by accuracy: 15

Final comparison: HIKNet vs SVM (eval set)

Metric	HIKNet	SVM
Accuracy	0.899	0.759
Precision	0.899	0.873
Specificity	0.899	0.911
Sensitivity	0.899	0.608
ROC AUC	0.950	0.910
PR AUC	0.923	0.902

Same held-out split, n = 158



Limitations

- Synthetic data only: no claim about real mouthguard performance
- Small dataset (527 events) -> wide variance across seeds and splits
- Single held-out split for the final comparison; no external test set
- SVM feature set re-selected per split, so its features differ between runs
- Mock generator may make the classes easier or harder than real data

Conclusion

- Full PyTorch pipeline: data, SVM baseline, HIKNet, CV, tuning, evaluation
- Eval split, HIKNet vs SVM: accuracy 0.899 vs 0.759, ROC AUC 0.950 vs 0.910
- Best tuned config: GAP head, 150 filters, k=15, dropout 0.5
- Next step: rerun on real instrumented-mouthguard data